

Step 1: Install Swagger Packages

Open terminal in your project folder and type:

```
npm install swagger-ui-express swagger-jsdoc
```

Step 2: Create swagger.js

Create a new file in the root folder

project/

|

|—— crudapi.js

|—— crudserver.js

|—— swagger.js ← New

|—— ...

Copy and paste the below code in swagger.js file

```
const swaggerJsDoc = require("swagger-jsdoc");

const options = {
  definition: {
    openapi: "3.0.0",
    info: {
      title: "CRUD API",
      version: "1.0.0",
      description: "Simple CRUD API using Express and MySQL"
    },
  },
  servers: [
    {
      url: "http://localhost:3000"
    }
  ]
}
```

```
},  
apis: ["/crudapi.js"]  
};  
module.exports = swaggerJsDoc(options);
```

Step 3: Modify crudserver.js

1. Add these two imports near the top:

```
const swaggerUi = require("swagger-ui-express");  
const swaggerSpec = require("./swagger");
```

2. Place this before app.listen():

```
app.use(  
  "/api-docs",  
  swaggerUi.serve,  
  swaggerUi.setup(swaggerSpec)  
);
```

Step 4: Document Your Routes

Since you mounted: `app.use("/crudapi", apiRoutes);`

all endpoints actually become:

- POST /crudapi/posting
- GET /crudapi/viewing
- PUT /crudapi/updating/{id}
- DELETE /crudapi/deleting/{id}
- GET /crudapi/testdb

Notice the /crudapi prefix.

Step 5: Add Swagger Comment Above POST Route

In crudapi.js, place the following comments immediately above:

```
router.post("/posting", async (req, res) => {
```

```
/**
```

```
* @swagger
```

```
* /crudapi/posting:
*   post:
*     summary: Insert a new record
*     tags:
*       - CRUD
*     requestBody:
*       required: true
*       content:
*         application/json:
*           schema:
*             type: object
*             properties:
*               name:
*                 type: string
*               age:
*                 type: integer
*               city:
*                 type: string
*     responses:
*       200:
*         description: Data inserted successfully
*/
```

Step 6: Above GET Route

```
/**
* @swagger
* /crudapi/viewing:
*   get:
*     summary: Get all records
```

```
* tags:
*   - CRUD
* responses:
*   200:
*     description: List of records
*/
```

Step 7: Above PUT Route

```
/**
* @swagger
* /crudapi/updating/{id}:
*   put:
*     summary: Update a record
*     tags:
*       - CRUD
*     parameters:
*       - in: path
*         name: id
*         required: true
*         schema:
*           type: integer
*     requestBody:
*       required: true
*       content:
*         application/json:
*           schema:
*             type: object
*             properties:
*               name:
```

```

*         type: string
*
*         age:
*
*         type: integer
*
*         city:
*
*         type: string
*
*   responses:
*
*     200:
*
*       description: Data updated successfully
*/

```

Step 8: Above DELETE Route

```

/**
* @swagger
* /crudapi/deleting/{id}:
*   delete:
*     summary: Delete a record
*     tags:
*       - CRUD
*     parameters:
*       - in: path
*         name: id
*         required: true
*         schema:
*           type: integer
*     responses:
*       200:
*         description: Data deleted successfully
*/

```

Step 9: Above Database Test Route

```
/**
 * @swagger
 * /crudapi/testdb:
 *   get:
 *     summary: Test database connection
 *     tags:
 *       - Database
 *     responses:
 *       200:
 *         description: Database connected successfully
 */
```

Step 10: Run the Server

Run: `node crudserver.js`

Open browser and type: <http://localhost:3000/api-docs>

You should see an interactive Swagger page with all 5 endpoints and a **Try it out** button for testing them.